

Illustration-42: Find the point of inflection of the function $y = x^3 - 3x^2 + 3x$.

Solution: Given $y = x^3 - 3x^2 + 3x \therefore \frac{dy}{dx} = 3x^2 - 6x + 3$

$$\text{Now } \frac{dy}{dx} = 0 \Rightarrow 3x^2 - 6x + 3 = 0 \Rightarrow 3(x-1)^2 = 0 \Rightarrow x = 1$$

Therefore, $\frac{dy}{dx} = 0$ when $x = 1$

$$\text{Thus } \frac{d^2y}{dx^2} = 6x - 6 \therefore \frac{d^2y}{dx^2} \Big|_{x=1} = 6 - 6 = 0$$

$$\text{Again } \frac{d^3y}{dx^3} = 6 \neq 0$$

$$\text{When } x = 1, y = 1^3 - 3 \times 1^2 + 3 \times 1 = 1$$

Point of inflection = (1,1) [Ans]

Illustration-43: Show that the function $y = xe^{-x}$ has a point of inflection at $x = 2$.

Solution: Given $y = xe^{-x} \therefore \frac{dy}{dx} = x(e^{-x})(-1) + e^{-x} = e^{-x}(1-x)$

$$\text{Now } \frac{d^2y}{dx^2} = e^{-x}(-1) + (1-x)(-1)e^{-x} = e^{-x}(x-2)$$

$$\frac{d^3y}{dx^3} = e^{-x}(1) + (x-2)(-1)e^{-x} = e^{-x}(3-x)$$

$$\text{For } x = 2, \text{ we have } \frac{d^2y}{dx^2} \Big|_{x=2} = e^{-2}(2-2) = 0 \text{ and } \frac{d^3y}{dx^3} \Big|_{x=2} = e^{-2}(3-2) = \frac{1}{e^2} \neq 0$$

Therefore $x = 2$ is the point of inflection. [Ans]

Illustration-44: Find the maximum and minimum values of the function: $x^3 - 2x^2 - 4x - 1$

Solution: Let $y = x^3 - 2x^2 - 4x - 1 \therefore \frac{dy}{dx} = 3x^2 - 4x - 4 \therefore \frac{d^2y}{dx^2} = 6x - 4$

For maximum or minimum, $\frac{dy}{dx} = 0$

$$\Rightarrow 3x^2 - 4x - 4 = 0 \Rightarrow 3x^2 - 6x + 2x - 4 = 0 \Rightarrow 3x(x-2) + 2(x-2) = 0$$

$$\Rightarrow (x-2)(3x+2) = 0 \Rightarrow x-2 = 0 \text{ or } 3x+2 = 0 \Rightarrow x = 2 \text{ or } x = -\frac{2}{3}$$

When $x = 2$, then $\frac{d^2y}{dx^2} = 6x - 4 = 6(2) - 4 = 8$, which is positive. So the function is minimum at $x = 2$.

$$\therefore \text{Minimum value } (y) = x^3 - 2x^2 - 4x - 1 = (2)^3 - 2(2)^2 - 4(2) - 1 = 8 - 8 - 8 - 1 = -9$$

When $x = -\frac{2}{3}$, then $\frac{d^2y}{dx^2} = 6x - 4 = 6\left(-\frac{2}{3}\right) - 4 = -8$, which is negative.

So the function is maximum at $x = -\frac{2}{3}$.

$$\begin{aligned} \therefore \text{Maximum value } (y) &= x^3 - 2x^2 - 4x - 1 = \left(-\frac{2}{3}\right)^3 - 2\left(-\frac{2}{3}\right)^2 - 4\left(-\frac{2}{3}\right) - 1 \\ &= -\frac{8}{27} - \frac{8}{9} + \frac{8}{3} - 1 = \frac{-8 - 24 + 72 - 27}{27} = \frac{13}{27} \end{aligned}$$

Answers: Maximum value $= \frac{13}{27}$ at $x = -\frac{2}{3}$, and Minimum value $= -9$ at $x = 2$

Illustration (45) If $y = x^4 - 3x^3 + 3x^2 + 5x + 1$, prove that $\frac{d^2y}{dx^2}$ is negative when

x lies between $\frac{1}{2}$ and 1 . What happens if $x = \frac{1}{2}$ or $x = 1$.

Solution: First Part:

$$\text{Given } y = x^4 - 3x^3 + 3x^2 + 5x + 1$$

$$\therefore \frac{dy}{dx} = 4x^3 - 9x^2 - 6x + 5$$

$$\therefore \frac{d^2y}{dx^2} = 12x^2 - 18x + 6$$

Let $x = \frac{3}{4}$, which lies between $\frac{1}{2}$ and 1 .

$$\frac{d^2y}{dx^2} = 12x^2 - 18x + 6 = 12\left(\frac{3}{4}\right)^2 - 18\left(\frac{3}{4}\right) + 6 = -0.75, \text{ which is negative. [Proved]}$$

Second Part:

When $x = \frac{1}{2}$ then $\frac{d^2y}{dx^2} = 12x^2 - 18x + 6 = 12\left(\frac{1}{2}\right)^2 - 18\left(\frac{1}{2}\right) + 6 = 3$, which is positive.

So the function is minimum at $x = \frac{1}{2}$. [Ans]

$$\text{When } x = 1, \text{ then } \frac{d^2y}{dx^2} = 12x^2 - 18x + 6$$

$$= 12(1)^2 - 18(1) + 6$$

$= 0$, which is neither positive nor negative.

Here $\frac{d^3y}{dx^3} = 24x - 18 = 24(1) - 18 = 6$, which is not zero.

So it has neither a maximum nor a minimum value at $x = 1$. It is a point of inflexion.

[Ans.]

Illustration-46: The difference of two numbers is 100. The square of the larger one exceeds five times the square of the smaller one by an amount which is maximum. Find the numbers.

Solution: Let the larger number $= x$

The smaller number $= y$

According to the question, $x - y = 100$

Also, let the maximum value $= h$

From the condition,

$$h = x^2 - 5y^2 \Rightarrow h = x^2 - 5(x - 100)^2 \Rightarrow h = x^2 - 5(x^2 - 200x + 10000)$$

$$\Rightarrow h = x^2 - 5x^2 + 1000x - 50000 \Rightarrow h = -4x^2 + 1000x - 50000$$

$$\therefore \frac{dh}{dx} = -8x + 1000 \quad \therefore \frac{d^2h}{dx^2} = -8$$

For maximum or minimum, $\frac{dh}{dx} = 0 \Rightarrow -8x + 1000 = 0$

$$\Rightarrow 8x^2 = 1000 \Rightarrow x = 125$$

When $x = 125$, then $\frac{d^2h}{dx^2} = -8$, which is negative. So the function is maximum at $x = 125$.

Hence h is maximum at $x = 125$.

$\therefore$ Required larger number $(x) = 125$

$\therefore$ Required smaller number $(y) = x - 100 = 125 - 100 = 25$

Answers: Larger number = 125, Smaller number = 25.

Illustration-47: Find the maximum or minimum value of $2x^3 - 15x^2 + 36x + 12$.

Solution: Let $y = 2x^3 - 15x^2 + 36x + 12$

$$\therefore \frac{dy}{dx} = 6x^2 - 30x + 36 \therefore \frac{d^2y}{dx^2} = 12x - 30$$

For maximum or minimum, $\frac{dy}{dx} = 0$

$$\Rightarrow 6x^2 - 30x + 36 = 0 \Rightarrow x^2 - 5x + 6 = 0 \Rightarrow x^2 - 3x - 2x + 6 = 0$$

$$\Rightarrow x(x - 3) - 2(x - 3) = 0 \Rightarrow (x - 3)(x - 2) = 0$$

$$\Rightarrow x-3=0 \text{ or } x-2=0 \quad \Rightarrow x=3 \text{ or } x=2$$

When $x=2$, then

$$\frac{d^2y}{dx^2} = 12x - 30 = 12(2) - 30 = -6 \text{ which is negative. So the function is maximum at } x=2.$$

$$\therefore \text{Required maximum value } (y) = 2x^3 - 15x^2 + 36x + 12 \\ = 2(2)^3 - 15(2)^2 + 36(2) + 12 = 16 - 60 + 72 + 12 = 40$$

When $x=3$, then

$$\frac{d^2y}{dx^2} = 12x - 30 = 12(3) - 30 = 6 \text{ which is positive. So the function is minimum at } x=3.$$

$$\therefore \text{Required maximum value } (y) = 2x^3 - 15x^2 + 36x + 12 \\ = 2(3)^3 - 15(3)^2 + 36(3) + 12 = 54 - 135 + 108 + 12 = 39$$

Answers: Max. value = 40 at $x=2$, and Min. value = 39 at $x=3$

Illustration 48 Prove that the curve given by $3y = x^3 - 3x^2 - 9x + 11$, has a maximum at $x=-1$, minimum at $x=3$, and point of inflexion when $x=1$.

Solution: (a) Let $3y = x^3 - 3x^2 - 9x + 11$

$$\Rightarrow y = \frac{x^3 - 3x^2 - 9x + 11}{3} \Rightarrow y = \frac{x^3}{3} - x^2 - 3x + \frac{11}{3}$$

$$\therefore \frac{dy}{dx} = x^2 - 2x - 3 \therefore \frac{d^2y}{dx^2} = 2x - 2$$

For maximum or minimum,

$$\frac{dy}{dx} = 0 \Rightarrow x^2 - 2x - 3 = 0 \Rightarrow x^2 - 3x + x - 3 = 0 \Rightarrow x(x-3) + (x-3) = 0$$

$$\Rightarrow (x-3)(x+1) = 0 \Rightarrow x-3=0 \text{ or } x+1=0 \Rightarrow x=3 \text{ or } x=-1$$

When $x=-1$, then

$$\frac{d^2y}{dx^2} = 2x - 2 = 2(-1) - 2 = -4, \text{ which is negative. So the function is maximum at } x=-1. \text{ [Proved]}$$

When $x=3$, then

$$\frac{d^2y}{dx^2} = 2x - 2 = 2(3) - 2 = 4, \text{ which is positive. So the function is minimum at } x=3. \text{ [Proved]}$$

When $x=1$, then

$$\frac{d^2y}{dx^2} = 2x - 2 = 2(1) - 2 = 0, \text{ which is neither positive nor negative.}$$

Here $\frac{d^3y}{dx^3} = 2$, which is not zero. So it has neither a maximum nor a minimum value at $x=1$.
It is a point of inflexion. [Proved]